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Types of Errors

Introduction to Computer Programming

Management and Artificial Intelligence

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Types of Errors in Python

We can differentiate errors in Python in three main categories:

1. Syntax Errors
2. Runtime Errors
3. Logical Errors

Syntax Errors

- They occur when Python cannot understand the code due to a violation of language rules.
- They happen **before** the program runs.
- If the syntax is wrong and the code is processed for execution, then Python will always detect it.
- They will be the most common type of error you will see!

If the syntax is not correct, Python cannot understand what the code is trying to do.

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If the syntax is not correct, Python cannot understand what the code is trying to do.

For instance, consider the following code:

```
print "Hello, World"
```

Can you spot the problem?

An example of syntax error

When attempting to run the code:

```
print "Hello, World"
```

```
Cell In[5], line 1
```

```
    print "Hello, World"
```

```
    ^
```

```
SyntaxError: Missing parentheses in call to 'print'. Did you mean print (...)?
```

An example of syntax error

When attempting to run the code:

```
print "Hello, World"
```

```
Cell In[5], line 1  
    print "Hello, World"  
    ^
```

```
SyntaxError: Missing parentheses in call to 'print'. Did you mean print  
(...)?
```

Python gives us a clear message about what went wrong:

1. The type of error: `SyntaxError`
2. The line number where the error occurred: line 1
3. The error message: Missing parentheses in call to 'print'
4. A suggestion: Did you mean `print(...)`?

WARNING: Read the error message carefully! It can save you a lot of time!

What is a possible fix to the code?

```
print("Hello, World!")
```

As suggested by the Python error message, we need to add the **parentheses** to the function call to print.

Runtime Errors

- They occur when an error happens during execution, causing the program to stop.
- They happen **while** the program is running.
- The error may happen only for some data values! You may not see the error for some data values. **This makes them tricky!**
- They will be the second most common type of error we will see!

Python can run your code but it stops because it detects an error.

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Python can run your code but it stops because it detects an error.

For instance, consider the following code:

```
x = 10  
y = 0  
z = x / y
```

Can you spot the problem?

An example of runtime error

```
x = 10  
y = 0  
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```

An example of runtime error

```
x = 10  
y = 0  
z = x / y
```

Python gives us a clear message about what went wrong:

1. The type of error: `ZeroDivisionError`
2. The line number where the error occurred: line 3
3. The error message: `division by zero`

WARNING: Read the error message carefully! It can save you a lot of time!

What is a possible fix to the code?

Two options:

1. We use a different value for the denominator y .

```
x = 10
y = 2
z = x / y
print("x is", x, "\ny is", y, "\nz is", z)
```

2. We check if the denominator is zero before performing the division. *We will see how to do this in the next lectures.*

Logical Errors

- The program runs but produces incorrect or unintended results.
- Happens **after** the program runs **without** errors.
- Since it does not raise an error, they are extremely **subtle** to find.
- To catch them, you need to reason about the **semantic** meaning of the code.

Logical Errors

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- Happens **after** the program runs **without** errors.
- Since it does not raise an error, they are extremely **subtle** to find.
- To catch them, you need to reason about the **semantic** meaning of the code.

For example, consider the following code:

```
a = 10
b = 2
sum_a_b = a - b
print("Sum of", a, "and", b, "is:", sum_a_b)
```

Can you spot the error?

Example of a logical error

```
a = 10  
b = 2  
sum_a_b = a - b  
print("Sum of", a, "and", b, "is:", sum_a_b)
```

- Python is not able to detect the error. **Why?**
- The error is about the semantic of your code.
- Python does not what is the end goal of your code.
- You are only knowing what is the end goal, thus the expected result.

What is a possible fix to the code?

```
a = 10  
b = 2  
sum_a_b = a + b  
print("Sum of", a, "and", b, "is:", sum_a_b)
```

We need to replace the operator - with + since our goal is to compute the sum of the two variables.

Additional examples

```
# syntax error  
print(3+2
```

Cell In[6], line 2

```
print(3+2
```

^

SyntaxError: incomplete input

FIX: close the parenthesis

```
print(3+2)
```

5

```
# syntax error  
print("hello world')
```

```
Cell In[9], line 2  
    print("hello world")  
      ^
```

SyntaxError: unterminated string literal (detected at line 2)

FIX: close the string with the same quote used to open it.

```
print("hello world")
```

```
# syntax error  
44_cats = "cartoon"
```

```
Cell In[10], line 2  
    44_cats = "cartoon"  
      ^
```

SyntaxError: invalid decimal literal

FIX: do not use numbers as prefix of variable names

```
cats_44 = "cartoon"
```

```
# runtime error  
print(my_var)
```

```
-----  
NameError                                Traceback (most recent call last)  
Cell In[11], line 2  
      1 # runtime error  
----> 2 print(my_var)  
  
NameError: name 'my_var' is not defined
```

FIX: define the variable before using it!

```
my_variable = 42  
print(my_variable)
```

42

```
# runtime error
temperature = "25°C"
temperature += 10 # Attempting to add an integer to a temperature string
print(temperature)
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[13], line 3
      1 # runtime error
      2 temperature = "25°C"
----> 3 temperature += 10 # Attempting to add an integer to a temperature
      string
      4 print(temperature)

TypeError: can only concatenate str (not "int") to str
```

FIX: keep temperature as an integer

```
temperature = 25
temperature += 10 # Attempting to add an integer to a temperature string
print(temperature, "°C")
```

35 °C

```
# runtime error
number = 42
text = "Hello, World!"
result = number + text
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[15], line 4
      2 number = 42
      3 text = "Hello, World!"
----> 4 result = number + text

TypeError: unsupported operand type(s) for +: 'int' and 'str'
```

FIX: convert number to string before concatenate it to text

```
number = 42
text = "Hello, World!"
result = str(number) + text
print(result)
```

42Hello, World!

```
# semantic error  
radius = 10  
area = 2 * 3.14159 * radius # Incorrect formula for calculating the area of a circle  
print(area)
```

62.8318

FIX: Use the proper formula $\pi * r^2$

```
radius = 10  
area = 3.14159 * (radius ** 2) # Correct formula for area of a circle  
print(area)
```

314.159

```
# semantic error  
price = 100  
discount = 20 # 20% discount  
final_price = price * (discount / 100) # Incorrect application of % discount  
print(final_price)
```

20.0

FIX: Use the correct formula!

```
price = 100  
discount = 20 # 20% discount  
final_price = price - (price * discount / 100) # Correct application of % discount  
print(final_price)
```

80.0

